Chapter 7 Principal Components Analysis

Content

§ 7.1 Introduction

- **Principal Component Analysis** （主成分分析, PCA） is introduced by Pearson(1901) and developed by Hotelling (1933).
- Principal component analysis is a kind of statistic analysis method to reduce multiple variables to a few principal components by dimensionality reduction. These principal components can reflect the vast majority of information of original variables and they are usually a certain linear combination of original variables.
- Objectives of PCA :

(1) reduce the dimensionality of variables ;

(2) interpret principal components 。

§ 7.1 Introduction

- Principal components :

  (1) they are uncorrelated ;

  (2) the first principal components accounts for as much of the variability in the data as possible

  (3) each succeeding components accounts for as much of the remaining variability as possible.

§ 7.2 Principal Components for population

- § 7.2.1 definition and derivation of principal components
- § 7.2.2 properties of principal components
- § 7.2.3 calculate principal components from correlation matrix

§ 7.2.1 definition and derivation of principal components

- Suppose $\boldsymbol{x}=(x_1,x_2,\cdots,x_p)'$ be a random p-vector, $E(\boldsymbol{x})=\boldsymbol{\mu}$, $V(\boldsymbol{x})=\boldsymbol{\Sigma}$. Consider the following linear transformation:

$$
\begin{aligned}
y_1 &= a_{11}x_1 + a_{21}x_2 + \cdots + a_{p1}x_p = \boldsymbol{a}_1'\boldsymbol{x} \\
y_2 &= a_{12}x_1 + a_{22}x_2 + \cdots + a_{p2}x_p = \boldsymbol{a}_2'\boldsymbol{x} \\
&\vdots \\
y_p &= a_{1p}x_1 + a_{2p}x_2 + \cdots + a_{pp}x_p = \boldsymbol{a}_p'\boldsymbol{x}
\end{aligned}
$$

Find a comprehensive variable y_1 to represents original p variables.

Seek for a vector $\boldsymbol{a}_1$ on the constraint condition $\boldsymbol{a}_1'\boldsymbol{a}_1 = 1$ to maximize

$$V(y_1) = \boldsymbol{a}_1'\boldsymbol{\Sigma}\boldsymbol{a}_1$$

y_1 is called First Principal Component.

- Let λ_i be the i-th eigenvalue of Σ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$, and t_i be the corresponding eigenvector. From spectral factorization (1.6.6),

$$\Sigma = T\Lambda T' = \sum_{i=1}^{p} \lambda_i t_i t_i'.$$

where $T = (t_1, t_2, \cdots, t_p) = (t_{ik})$ is an orthogonal matrix and the diagonal matrix

$$\Lambda = diag(\lambda_1, \lambda_2, \cdots, \lambda_p).$$

- It can be verified from (1.8.3) that

$$a_1'\Sigma a_1 \leq \lambda_1$$

if and only if $a_1 = t_1$.

Therefore, $y_1 = t_1'x$ is the first principal component and its variance has the maximum λ_1.

(4) If A is p dimensional symmetric matrix and its eigenvalues are $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$, then

$$\max_{x \neq 0} \boldsymbol{x}' \boldsymbol{A} \boldsymbol{x} / \boldsymbol{x}' \boldsymbol{x} = \lambda_1 \qquad (1.8.3)$$

$$\min_{x \neq 0} \boldsymbol{x}' \boldsymbol{A} \boldsymbol{x} / \boldsymbol{x}' \boldsymbol{x} = \lambda_p \qquad (1.8.4)$$

The equality holds when $x = t_1$ and $x = t_p$, respectively.

- If the information of first principal component is not enough to represent original p vectors, then consider another aggregate variable $y_2 = \boldsymbol{a}_2'\boldsymbol{x}$. In order to underlap(不重叠) the information of y_2 and y_1, we need to use the condition

$$\mathrm{Cov}(y_1, y_2) = 0$$

Seek for vector $\boldsymbol{a}_2$ on this constraint condition $\boldsymbol{a}_2'\boldsymbol{a}_2 = 1$ in order to maximize $V(y_2) = \boldsymbol{a}_2'\boldsymbol{\Sigma}\boldsymbol{a}_2$. Then y_2 is called Second Principal Component:

$$y_2 = t_{12}x_1 + t_{22}x_2 + \cdots + t_{p2}x_p = \boldsymbol{t}_2'\boldsymbol{x}$$

its variance is λ_2.

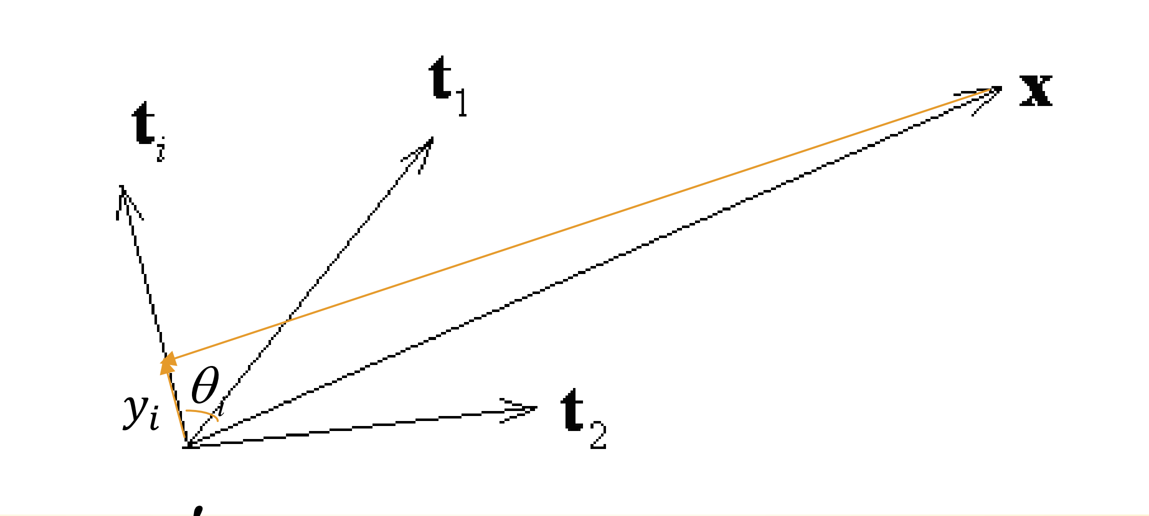
- Generally speaking, the i-th principal component of $\boldsymbol{x}$ is: seek for $\boldsymbol{a}_i$ under the conditions of $\boldsymbol{a}_i'\boldsymbol{a}_i = 1$ and Cov(y_k,y_i)=0, k=1,2,⋯,i−1 in order to maximize $V(y_i) = \boldsymbol{a}_i'\boldsymbol{\Sigma}\boldsymbol{a}_i$.

From spectral factorization (1.6.6), the i-th principal component is

$$y_i = t_{1i}x_1 + t_{2i}x_2 + \cdots + t_{pi}x_p = \boldsymbol{t}_i'\boldsymbol{x}, \quad i = 1, 2, \cdots, p$$

Geometric significance of principal components

- In geometrics, $\boldsymbol{t}_i$ represents the i-th principal component's direction. y_i is the projection of $\boldsymbol{x}$ on $\boldsymbol{t}_i$ (its absolute value is projected length). λ_i is the variance of these projection values. It reflects the degree of dispersion of projection points on $\boldsymbol{t}_i$.



$$\|\boldsymbol{x}\|\cos\theta_i = \|\boldsymbol{x}\|\frac{\boldsymbol{t}_i'\boldsymbol{x}}{\|\boldsymbol{t}_i\|\|\boldsymbol{x}\|} = \boldsymbol{t}_i'\boldsymbol{x} = y_i \text{ when } \|t_i\|=1$$

Projection of $\boldsymbol{x}$ on $\boldsymbol{t}_i$

- $\|\boldsymbol{x}\|\cos\theta_i = \|\boldsymbol{x}\|\dfrac{\boldsymbol{t}_i'\boldsymbol{x}}{\|\boldsymbol{t}_i\|\|\boldsymbol{x}\|} = \boldsymbol{t}_i'\boldsymbol{x} = y_i$ when $\|t_i\|=1$

where θ_i is the included angle of $\boldsymbol{t}_i$ and $\boldsymbol{x}$.

Relationship between principal components and original variables:

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix} = \begin{pmatrix} \boldsymbol{t}_1'\boldsymbol{x} \\ \boldsymbol{t}_2'\boldsymbol{x} \\ \vdots \\ \boldsymbol{t}_p'\boldsymbol{x} \end{pmatrix} = \begin{pmatrix} \boldsymbol{t}_1' \\ \boldsymbol{t}_2' \\ \vdots \\ \boldsymbol{t}_p' \end{pmatrix} \boldsymbol{x} = \boldsymbol{T}'\boldsymbol{x}$$

where $\boldsymbol{T} = \left(\boldsymbol{t}_1, \boldsymbol{t}_2, \cdots, \boldsymbol{t}_p\right) = \left(t_{ik}\right)$ is the orthogonal matrix.

$$\boldsymbol{x} = \boldsymbol{T}\boldsymbol{y}$$

$$
\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix} = \begin{pmatrix} t_{11} & t_{21} & \cdots & t_{p1} \\ t_{12} & t_{22} & \cdots & t_{p2} \\ \vdots & \vdots & & \vdots \\ t_{1p} & t_{2p} & \cdots & t_{pp} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}
$$

$$
\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} t_{11} & t_{12} & \cdots & t_{1p} \\ t_{21} & t_{22} & \cdots & t_{2p} \\ \vdots & \vdots & & \vdots \\ t_{p1} & t_{p2} & \cdots & t_{pp} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix}
$$

$$
\boldsymbol{x} = \boldsymbol{T}\boldsymbol{y}
$$

Relationship Matrix between principal components and original variables:

	y_1	y_2	$\cdots$	y_p
x_1	t_{11}	t_{12}	$\cdots$	t_{1p}
x_2	t_{21}	t_{22}	$\cdots$	t_{2p}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
x_p	t_{p1}	t_{p2}	$\cdots$	t_{pp}
	$\boldsymbol{t}_1$	$\boldsymbol{t}_2$		$\boldsymbol{t}_p$

Geometric Meaning of Orthogonal Transformation of $\boldsymbol{y} = \boldsymbol{T}'\boldsymbol{x}$

- Geometric meaning of orthogonal transformation of $\boldsymbol{y} = \boldsymbol{T}'\boldsymbol{x}$: take an orthogonal rotation of original p-dimensional coordinate axis consisting of $x_1, x_2, \cdots, x_p$ in R^p. One group of orthogonal unit vectors $y_1, y_2, \cdots, y_p$ represent p directions of new coordinate axises, which are still mutually orthogonal.

Summary: calculate principal component from Σ

- Procedures:

step 1: Find eigenvalues of Σ $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$ and the corresponding mutually orthogonal unit eigenvectors $t_1, t_2, \cdots, t_p$.

Step 2: Construct the i-th population principal component $y_i = t_i' x$ which has the variance $\lambda_i, i = 1,2,\cdots,p$, and

$$Cov(y_i, y_j) = 0 (i \neq j).$$

Outputs: p population principal components: $y_1 = t_1' x, y_2 = t_2' x, \cdots, y_p = t_p' x$.

Notes: In geometry, orientations of p principal are the directions of $t_1, t_2, \cdots, t_p$. Projection points on t_1 of n samples are most dispersal. The dispersion degree of projection points of n samples on $t_2, \cdots, t_p$ is decreasing successively.

§ 7.2.2 properties of principal components

- 1. covariance matrix of principal components
- 2. total variance of principal components
- 3. correlation coefficient of original variables x_i and principal component y_k
- 4. Rate of contribution of m principal components to original variables
- 5. Influence of original variables to principal components

1. covariance matrix of principal components

$$V(\boldsymbol{y})=\boldsymbol{\Lambda}$$

where $\boldsymbol{\Lambda}$=diag(λ_1, λ_2,$\cdots$,λ_p), i.s.:$V(y_i)= \lambda_i$, i=1,2,$\cdots$,p and the principal components y_1,y_2, $\cdots$,y_p are mutually uncorrelated.

2. total variance of principal components

- Since

$$\operatorname{tr}(\Lambda) = \operatorname{tr}\left(\boldsymbol{T}'\boldsymbol{\Sigma}\boldsymbol{T}\right) = \operatorname{tr}\left(\boldsymbol{\Sigma}\boldsymbol{T}\boldsymbol{T}'\right) = \operatorname{tr}\left(\boldsymbol{\Sigma}\right)$$

then

$$\sum_{i=1}^{p} \lambda_i = \sum_{i=1}^{p} \sigma_{ii}$$

or

$$\sum_{i=1}^{p} V\left(y_i\right) = \sum_{i=1}^{p} V\left(x_i\right)$$

- The ratio belonging to the ith principal component y_i (or explained by y_i) in total variance is

$$\lambda_i \Big/ \sum_{i=1}^{p} \lambda_i \qquad (7.2.7)$$

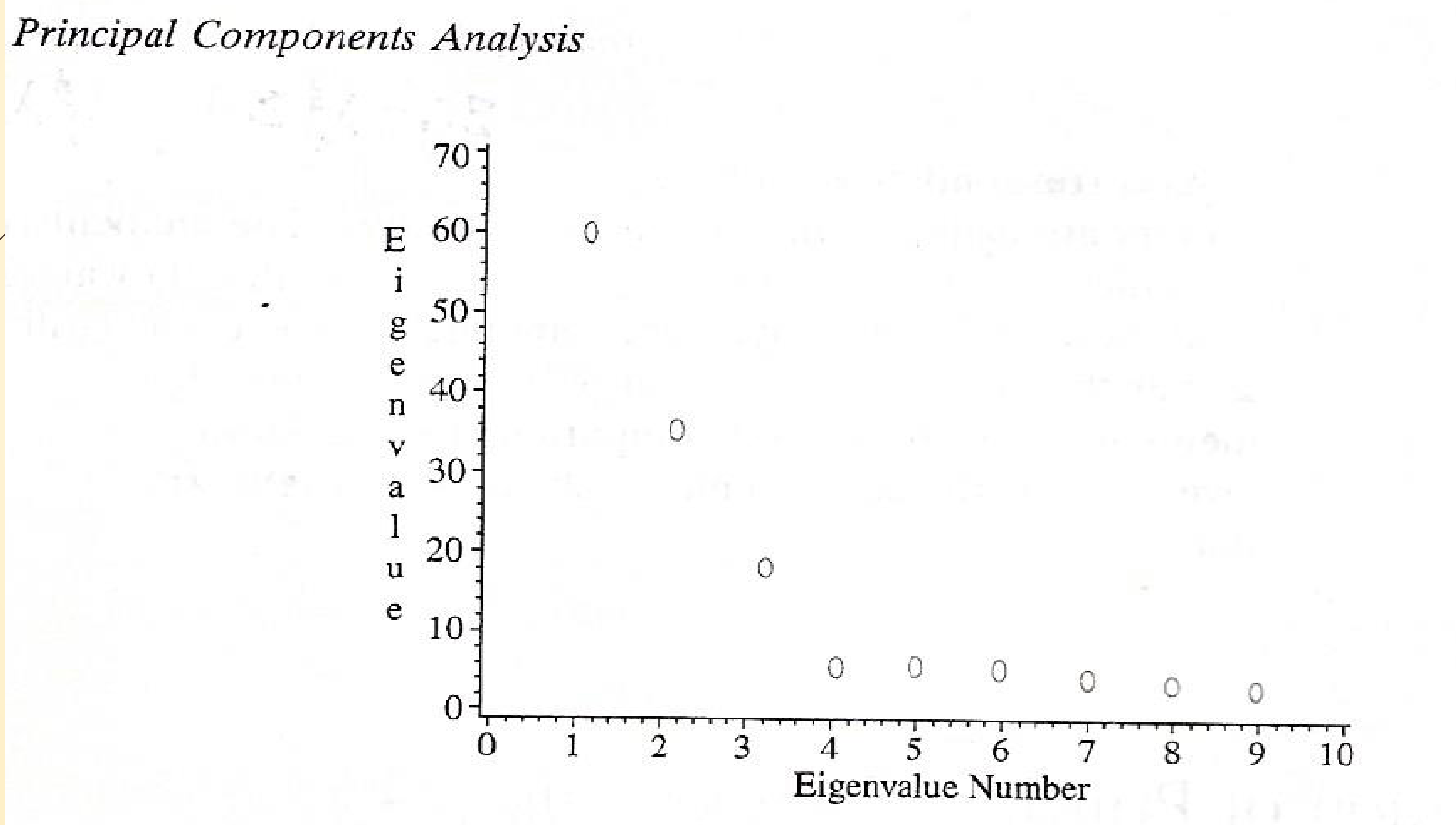
which is called the rate of construction of y_i.
- The rate of construction of first principal component y_1 is the maximum. It indicates that it has the strongest ability of explaining original variables $x_1, x_2, \cdots, x_p$. Besides, the explanation ability of $y_2, y_3, \cdots, y_p$ decrease in turn.
- Purpose of principal component analysis is to reduce the number of variables. Therefore, not all of p principal components are used. Ignoring some principal components with smaller variance will not lead to much influence of total variance.

- To choose the number of principal components:
- Method 1： Accumulative Rate of Construction of the first m principal components

$$\sum_{i=1}^{m}\lambda_i \Big/ \sum_{i=1}^{p}\lambda_i \qquad (7.2.8)$$

which represents the ability of $y_1,y_2, \cdots,y_m$ explaining $x_1,x_2, \cdots,x_p$.
- Usually, only m (compared to p) principal components are needed if our aim is to maintain the accumulative rate of construction to reach such a percentage (80%-90). Then, $x_1,x_2, \cdots,x_p$ can be replaced by $y_1,y_2, \cdots,y_m$ to reduce dimension with little lost of information.

- To choose the number of principal components:
- Method 2: A screen plot of rate of construction for each eigenvalue



3.Correlation coefficients between original variable x_i and principal component y_k

- $$\boldsymbol{x}=\boldsymbol{Ty} \qquad (7.2.9)$$

  i.e.,

  $$x_i=t_{i1}y_1+t_{i2}y_2+\ldots+t_{ip}y_p \qquad (7.2.9')$$

  so

  $$\mathrm{Cov}(x_i,y_k)=\mathrm{Cov}(t_{ik}y_k,\ y_k)=t_{ik}\lambda_k$$

- In real application, we are usually only interested in the correlation coefficients of x_i $(i=1,2,\cdots,p)$ and y_k $(k=1,2,\cdots,m)$.

$$\rho\left(x_i,y_k\right)=\frac{\mathrm{Cov}\left(x_i,y_k\right)}{\sqrt{V\left(x_i\right)}\sqrt{V\left(y_k\right)}}=\frac{\sqrt{\lambda_k}}{\sqrt{\sigma_{ii}}}t_{ik}, \quad i,k=1,2,\ldots,p \qquad (7.2.10)$$

4.Contribution rate of the *m* principal components to original variables

- The information of original variables x_i contained in *m* principal components $y_1,y_2,\cdots,y_m$ can be measured by the square of multiple correlation coefficients of x_i and $y_1,y_2,\cdots,y_m$, which is called **Rate of Contribution** of *m* principal components $y_1,y_2,\cdots,y_m$ to original variables x_i. Its value is:

$$\rho^2_{i\cdot 1,\cdots,m}=\frac{\left(\mathrm{Cov}(x_i,y_1),\cdots,\mathrm{Cov}(x_i,y_m)\right)\begin{pmatrix}V^{-1}(y_1) & & 0\\ & \ddots & \\ 0 & & V^{-1}(y_m)\end{pmatrix}\begin{pmatrix}\mathrm{Cov}(x_i,y_1)\\ \vdots\\ \mathrm{Cov}(x_i,y_m)\end{pmatrix}}{V(x_i)} \qquad (3.4.1)$$

$$=\left(\rho(x_i,y_1),\cdots,\rho(x_i,y_m)\right)\begin{pmatrix}\rho(x_i,y_1)\\ \vdots\\ \rho(x_i,y_m)\end{pmatrix}=\sum_{k=1}^{m}\rho^2(x_i,y_k)=\sum_{k=1}^{m}\lambda_k t_{ik}^2/\sigma_{ii} \qquad (7.2.11)$$

- When $m=p$, $\sum_{k=1}^{p}\rho^2(x_i,y_k)=\sum_{k=1}^{p}\lambda_k t_{ik}^2/\sigma_{ii}=1$ by (7.2.9). (7.2.12)

Example 7.2.1 Suppose the covariance matrix of $\boldsymbol{x}=(x_1,x_2,x_3)'$ is

$$\boldsymbol{\Sigma}=\begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

its eigenvalue is

$$\lambda_1=5.83,\quad \lambda_2=2.00,\quad \lambda_3=0.17$$

corresponding eigenvectors:

$$\boldsymbol{t}_1=\begin{pmatrix} 0.383 \\ -0.924 \\ 0.000 \end{pmatrix},\quad \boldsymbol{t}_2=\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},\quad \boldsymbol{t}_3=\begin{pmatrix} 0.924 \\ 0.383 \\ 0.000 \end{pmatrix}$$

If only one principal component is used, then rate of contribution:

$$5.83/(5.83+2.00+0.17)=0.72875=72.875\%$$

Table 7.2.1 **rate of contribution of y_1 and (y_1, y_2) to each original variables**

i	$\rho(y_1, x_i)$	$\rho^2_{i\cdot 1}$	$\rho(y_2, x_i)$	$\rho^2_{i\cdot 1,2} = \sum_{k=1}^{2} \rho^2(x_i, y_k)$
1	0.925	0.855	0.000	0.855
2	−0.998	0.996	0.000	0.996
3	0.000	0.000	1.000	1.000

$$\rho(x_i, y_k) = \frac{\sqrt{\lambda_k}}{\sqrt{\sigma_{ii}}} t_{ik}$$

$$\rho(x_1, y_1) = \frac{\sqrt{\lambda_1}}{\sqrt{\sigma_{11}}} t_{11} = \frac{\sqrt{5.83}}{\sqrt{1}} \times 0.383 = 0.9249 \qquad \rho(x_2, y_1) = \frac{\sqrt{\lambda_1}}{\sqrt{\sigma_{22}}} t_{21} = \frac{\sqrt{5.83}}{\sqrt{5}} \times (-0.924) = -0.9979$$

Rate of contribution of y_1 to third variable is zero. Because x_3 doesn't relate to x_1 and x_2. y_1 doesn't contain any information about x_3. So, y_2 should be used. Now, the rate of contribution is:

$$(5.83+2.00)/8=97.875\%$$

rate of contribution of (y_1, y_2) to every variable x_i are respectively

$\rho^2_{1\cdot 1,2} = 85.5\%$, $\rho^2_{2\cdot 1,2} = 99.6\%$, $\rho^2_{3\cdot 1,2} = 100\%$ are higher.

5.Effects of original variables on principal components

- $y_k=t_{1k}x_1+t_{2k}x_2+\cdots+t_{pk}x_p$. t_{ik} is **loading (载荷)** of kth principal component y_k to ith original variable x_i. It reflects the importance of x_i to y_k.
- For interpreting principal components, we need to examine the loadings and also consider the correlation coefficients. From equation (7.2.10), it is known that the correlation coefficients $\rho(x_i, y_k)$ have the same sign as the loadings t_{ik} and are directly proportional to them.
- Notice that $x_i=t_{i1}y_1+t_{i2}y_2+\ldots+t_{ip}y_p$, so $\sigma_{ii}=t_{i1}^2\lambda_1+t_{i2}^2\lambda_2+\cdots+t_{ip}^2\lambda_p$ is the weighted average of $\lambda_1,\lambda_2,\cdots,\lambda_p$.
- So variables with big variance has close relationship with principal components with big eigenvalues and vice versa.

$$\rho(x_i,y_k)=\frac{\mathrm{Cov}(x_i,y_k)}{\sqrt{V(x_i)}\sqrt{V(y_k)}}=\frac{\sqrt{\lambda_k}}{\sqrt{\sigma_{ii}}}t_{ik},\quad i,k=1,2,\ldots,p \qquad (7.2.10)$$

Example 7.2.2 Suppose $\boldsymbol{x}=(x_1,x_2,x_3)'$'s covariance matrix

$$\boldsymbol{\Sigma}=\begin{pmatrix}16 & 2 & 30\\ 2 & 1 & 4\\ 30 & 4 & 100\end{pmatrix}$$

After calculation, the eigenvalues and eigenvectors of $\boldsymbol{\Sigma}$

$$\lambda_1=109.793,\quad \lambda_2=6.469,\quad \lambda_3=0.738$$

$$\boldsymbol{t}_1=\begin{pmatrix}0.305\\ 0.041\\ 0.951\end{pmatrix},\quad \boldsymbol{t}_2=\begin{pmatrix}0.944\\ 0.120\\ -0.308\end{pmatrix},\quad \boldsymbol{t}_3=\begin{pmatrix}-0.127\\ 0.992\\ -0.002\end{pmatrix}$$

The corresponding principal components are

$$y_1=0.305x_1+0.041x_2+0.951x_3$$

$$y_2=0.944x_1+0.120x_2-0.308x_3$$

$$y_3=-0.127x_1+0.992x_2-0.002x_3$$

x_3 with big variance to a great extent controls first principal component y_1, while x_2 with small variance almost completely controls third principal component y_3. x_1 with intermediate variance basically controls second principal component y_2. Rate of contribution of y_1:

$$\frac{\lambda_1}{\lambda_1+\lambda_2+\lambda_3}=\frac{109.793}{117}=0.938\cdot$$

The high contribution rate primarily stems from the much larger variance of x_1 compared to x_2 and x_3, and secondarily from the intercorrelation among x_1, x_2 and x_3. The eigenvalue of y_3 is small. This reflects there exists a linear relationship of x_1,x_2,x_3 such that

$$-0.127x_1+0.992x_2-0.002x_3\approx c$$

where $c=-0.127\mu_1+0.992\mu_2-0.002\mu_3$ is a constant.

Summary: calculate principal component from Σ

- <u>Procedures</u>:

step 1: Find eigenvalues of Σ $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$ and the corresponding

mutually orthogonal unit eigenvectors $t_1, t_2, \cdots, t_p$.

Step 2: Construct the i-th population principal component $y_i = t_i' x$ which has the variance $\lambda_i, i = 1,2,\cdots,p$, and

$$Cov(y_i, y_j) = 0 (i \neq j).$$

Outputs: p population principal components: $y_1 = t_1' x, y_2 = t_2' x, \cdots, y_p = t_p' x$.

Notes: In geometry, orientations of p principal are the directions of $t_1, t_2, \cdots, t_p$. Projection points on t_1 of n samples are most dispersal. The dispersion degree of projection points of n samples on $t_2, \cdots, t_p$ is decreasing successively.

§ 7.2.3 calculate principal components from correlation matrix

- Common standard transformation is $x_i^* = \dfrac{x_i - \mu_i}{\sqrt{\sigma_{ii}}}, \quad i = 1, 2, \cdots, p$.
- Obviously, covariance matrix of $\boldsymbol{x}^* = \left(x_1^*, x_2^*, \cdots, x_p^*\right)'$ is the correlation matrix of $\boldsymbol{x}$, $\boldsymbol{R}$ **(2.2.21)** .
- Calculate principal components from $\boldsymbol{R}$. Each original variable is equally treated by doing principal component.
- The methods to calculate principal components from $\boldsymbol{R}$ and $\boldsymbol{\Sigma}$ are exactly the same. There exist simple formations of properties of principal components. Suppose $\lambda_1^* \geq \lambda_2^* \geq \cdots \geq \lambda_p^* \geq 0$ are p eigenvalues of $\boldsymbol{R}$ and $\boldsymbol{t}_1^*, \boldsymbol{t}_2^*, \cdots, \boldsymbol{t}_p^*$ are corresponding unit eigenvectors and they are mutually orthogonal. Then, p principal components are $\boldsymbol{y}^* = \left(y_1^*, y_2^*, \cdots, y_p^*\right)'$, $T^* = \left(\boldsymbol{t}_1^*, \boldsymbol{t}_2^*, \cdots, \boldsymbol{t}_p^*\right) = \left(t_{ik}^*\right)$. Suppose

  $y_1^* = \boldsymbol{t}_1^{*\prime} \boldsymbol{x}^*, y_2^* = \boldsymbol{t}_2^{*\prime} \boldsymbol{x}^*, \cdots, y_p^* = \boldsymbol{t}_p^{*\prime} \boldsymbol{x}^*$, then $\boldsymbol{y}^* = \boldsymbol{T}^{*\prime} \boldsymbol{x}^*$.

Principal Component Properties from $\boldsymbol{R}$

- (1) $E(\boldsymbol{y}^*)=\boldsymbol{0}$, $V(\boldsymbol{y}^*)=\boldsymbol{\Lambda}^*$, where $\boldsymbol{\Lambda}^* = \mathrm{diag}\left(\lambda_1^*, \lambda_2^*, \cdots, \lambda_p^*\right)$
- (2) $\sum_{i=1}^{p} \lambda_i^* = p$
- (3) Correlation coefficient between x_i^* and principal component y_k^* :

$$\rho\left(x_i^*,\ y_k^*\right) = \sqrt{\lambda_k^*}\, t_{ik}^*,\quad i,k = 1,2,\ldots,p\,. \qquad (7.2.17)$$

This leads to

$$\rho\left(x_i, y_k\right) = \frac{\mathrm{Cov}(x_i, y_k)}{\sqrt{V\left(x_i\right)}\sqrt{V\left(y_k\right)}} = \frac{\sqrt{\lambda_k}}{\sqrt{\sigma_{ii}}}\, t_{ik},\quad i,k = 1,2,\ldots,p \qquad (7.2.10)$$

$$\frac{\rho\left(x_1^*,\ y_k^*\right)}{t_{1k}^*} = \frac{\rho\left(x_2^*,\ y_k^*\right)}{t_{2k}^*} = \cdots = \frac{\rho\left(x_p^*,\ y_k^*\right)}{t_{pk}^*} = \sqrt{\lambda_k^*}$$

Therefore, when explaining principal component y_k^* , we can conclude that the effect of factor $t_{1k}^*, t_{2k}^*, \cdots, t_{pk}^*$ and correlation coefficient $\rho(x_1^*,\ y_k^*), \rho(x_2^*,\ y_k^*), \cdots,$ $\rho(x_p^*,\ y_k^*)$ are the same by noticing that $x_i^* = t_{i1}^* y_1^* + t_{i2}^* y_2^* + \cdots + t_{ip}^* y_p^*$.

So, we only need choose one kind of index to explain.

- (4) Rate of contribution of $y_1^*, y_2^*, \cdots, y_m^*$ to x_i^* from (7.2.11):

$$\rho_{i\bullet 1,\cdots,m}^2 = \sum_{k=1}^{m} \rho^2(x_i^*,\ y_k^*) = \sum_{k=1}^{m} \lambda_k^* t_{ik}^{*2}$$

- (5) $\sum_{k=1}^{p} \rho^2(x_i^*,\ y_k^*) = \sum_{k=1}^{p} \lambda_k^* t_{ik}^{*2} = 1$ from (7.2.12) or (7.2.9’).

Example 7.2.3 In example 7.2.2, correlation matrix of ***x:***

$$\boldsymbol{R}=\begin{pmatrix} 1 & 0.5 & 0.75 \\ 0.5 & 1 & 0.4 \\ 0.75 & 0.4 & 1 \end{pmatrix}$$

R's eigenvalues and eigenvectors are:

$$\lambda_1^* = 2.114, \quad \lambda_2^* = 0.646, \quad \lambda_3^* = 0.240$$

$$\boldsymbol{t}_1^* = \begin{pmatrix} 0.627 \\ 0.497 \\ 0.600 \end{pmatrix}, \quad \boldsymbol{t}_2^* = \begin{pmatrix} -0.241 \\ 0.856 \\ -0.457 \end{pmatrix}, \quad \boldsymbol{t}_3^* = \begin{pmatrix} -0.741 \\ 0.142 \\ 0.656 \end{pmatrix}$$

Corresponding principal components:

$$y_1^* = 0.627x_1^* + 0.497x_2^* + 0.600x_3^*$$

$$y_2^* = -0.241x_1^* + 0.856x_2^* - 0.457x_3^*$$

$$y_3^* = -0.741x_1^* + 0.142x_2^* + 0.656x_3^*$$

y_1^*'s rate of contribution:

$$\frac{\lambda_1^*}{3} = \frac{2.114}{3} = 0.705$$

Rate of accumulative contribution y_1^* and y_2^*

$$\frac{\lambda_1^* + \lambda_2^*}{3} = \frac{2.114 + 0.646}{3} = 0.920$$

- Now compare the results of calculating principal components from ***R*** and ***Σ***. Rate of contribution of y_1^* from ***R***, 0.705, is smaller than that of y_1 from ***Σ***, 0.938. In fact, the bigger difference between original variables' variances, the more obvious the trend is.
- y_1^*, y_2^*, y_3^* can be expressed by unstandardized original variables:

$$y_1^* = 0.627\left(\frac{x_1-\mu_1}{4}\right) + 0.497\left(\frac{x_2-\mu_2}{1}\right) + 0.600\left(\frac{x_3-\mu_3}{10}\right)$$

$$= 0.157(x_1-\mu_1) + 0.497(x_2-\mu_2) + 0.060(x_3-\mu_3)$$

$$y_2^* = -0.241\left(\frac{x_1-\mu_1}{4}\right)+0.856\left(\frac{x_2-\mu_2}{1}\right)-0.457\left(\frac{x_3-\mu_3}{10}\right)$$
$$= -0.060(x_1-\mu_1)+0.856(x_2-\mu_2)-0.046(x_3-\mu_3)$$

$$y_3^* = -0.741\left(\frac{x_1-\mu_1}{4}\right)+0.142\left(\frac{x_2-\mu_2}{1}\right)+0.656\left(\frac{x_3-\mu_3}{10}\right)$$
$$= -0.185(x_1-\mu_1)+0.142(x_2-\mu_2)+0.066(x_3-\mu_3)$$

- The relative magnitudes of loadings of y_i on the original variables are very different from the relative magnitudes of loadings of y_i on x_1, x_2, and x_3 as shown in Example 7.2.2. After standardization, the conclusion may be completely changed. Thus, standardization is not inessential.

Assignment： See the assignment file in BB

§ 7.3 Principal Components of Samples

- We can obtain principal components from $\boldsymbol{\Sigma}$ or $\boldsymbol{R}$. But in real problems, $\boldsymbol{\Sigma}$ or $\boldsymbol{R}$ are usually unknown and obtained by estimating samples. Suppose data matrix is

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1' \\ \boldsymbol{x}_2' \\ \vdots \\ \boldsymbol{x}_n' \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

Then sample covariance matrix and sample correlation coefficient are:

$$\boldsymbol{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}})(\boldsymbol{x}_i - \overline{\boldsymbol{x}})' = \left(s_{ij}\right)$$

$$\hat{\boldsymbol{R}} = \left(r_{ij}\right), \quad r_{ij} = \frac{s_{ij}}{\sqrt{s_{ii}}\sqrt{s_{jj}}}$$

§ 7.3 Principal Components of Samples

- § 7.3.1 definition of sample principal components
- § 7.3.2 calculate principal component from $\boldsymbol{S}$
- § 7.3.3 calculate principal component from $\hat{\boldsymbol{R}}$
- § 7.3.4 application of principal components
- § 7.3.5 supplement and attentions

§ 7.3.1 definition of sample principal components

- If variable $\boldsymbol{a}_1$ leads to the maximum of sample variance

$$\frac{1}{n-1}\sum_{j=1}^{n}\left(\boldsymbol{a}_1'\boldsymbol{x}_j-\boldsymbol{a}_1'\bar{\boldsymbol{x}}\right)^2$$

$$=\frac{1}{n-1}\sum_{j=1}^{n}\boldsymbol{a}_1'\left(\boldsymbol{x}_j-\bar{\boldsymbol{x}}\right)\left(\boldsymbol{x}_j-\bar{\boldsymbol{x}}\right)'\boldsymbol{a}_1=\boldsymbol{a}_1'\boldsymbol{S}\boldsymbol{a}_1$$

under the condition $\boldsymbol{a}_1'\boldsymbol{a}_1=1$, then linear combination $\hat{y}_1=\boldsymbol{a}_1'\boldsymbol{x}$ is called First Sample Principal Component.

- On the constraint conditions of $\boldsymbol{a}_2'\boldsymbol{a}_2=1$ and the sample covariance of $\left(\boldsymbol{a}_1'\boldsymbol{x}_1,\boldsymbol{a}_2'\boldsymbol{x}_1\right),\left(\boldsymbol{a}_1'\boldsymbol{x}_2,\boldsymbol{a}_2'\boldsymbol{x}_2\right),\cdots,\left(\boldsymbol{a}_1'\boldsymbol{x}_n,\boldsymbol{a}_2'\boldsymbol{x}_n\right)$ is

$$\frac{1}{n-1}\sum_{j=1}^{n}\left(\boldsymbol{a}_1'\boldsymbol{x}_j-\boldsymbol{a}_1'\bar{\boldsymbol{x}}\right)\left(\boldsymbol{a}_2'\boldsymbol{x}_j-\boldsymbol{a}_2'\bar{\boldsymbol{x}}\right) \qquad \operatorname{cov}(a_1'x,a_2'x)=a_1'\Sigma a_2=0$$

$$=\frac{1}{n-1}\sum_{j=1}^{n}\boldsymbol{a}_1'\left(\boldsymbol{x}_j-\bar{\boldsymbol{x}}\right)\left(\boldsymbol{x}_j-\bar{\boldsymbol{x}}\right)'\boldsymbol{a}_2=\boldsymbol{a}_1'\boldsymbol{S}\boldsymbol{a}_2=0$$

approximate

If variable $\boldsymbol{a}_2$ leads to the maximum of the sample variance of $\boldsymbol{a}_2'\boldsymbol{x}_1, \boldsymbol{a}_2'\boldsymbol{x}_2, \cdots, \boldsymbol{a}_2'\boldsymbol{x}_n$

$$\frac{1}{n-1}\sum_{j=1}^{n}\left(\boldsymbol{a}_2'\boldsymbol{x}_j - \boldsymbol{a}_2'\overline{\boldsymbol{x}}\right)^2 = \boldsymbol{a}_2'\boldsymbol{S}\boldsymbol{a}_2$$

then linear combination $\hat{y}_2 = \boldsymbol{a}_2'\boldsymbol{x}$ is called Second Sample Principal Component.

- Generally, on the constraint conditions of $\boldsymbol{a}_i'\boldsymbol{a}_i = 1$ and the sample covariance of $(\boldsymbol{a}_k'\boldsymbol{x}_1, \boldsymbol{a}_i'\boldsymbol{x}_1), (\boldsymbol{a}_k'\boldsymbol{x}_2, \boldsymbol{a}_i'\boldsymbol{x}_2), \cdots, (\boldsymbol{a}_k'\boldsymbol{x}_n, \boldsymbol{a}_i'\boldsymbol{x}_n)$ is

$$\operatorname{cov}(a_k'x, a_i'x) = a_k'\Sigma a_i = 0$$

approximate

$$\frac{1}{n-1}\sum_{j=1}^{n}\left(\boldsymbol{a}_k'\boldsymbol{x}_j - \boldsymbol{a}_k'\overline{\boldsymbol{x}}\right)\left(\boldsymbol{a}_i'\boldsymbol{x}_j - \boldsymbol{a}_i'\overline{\boldsymbol{x}}\right) = \boldsymbol{a}_k'\boldsymbol{S}\boldsymbol{a}_i = 0, \quad k = 1, 2, \cdots, i-1$$

If variable $\boldsymbol{a}_i$ leads to the maximum of the sample variance of

$$\frac{1}{n-1}\sum_{j=1}^{n}\left(\boldsymbol{a}_i'\boldsymbol{x}_j-\boldsymbol{a}_i'\overline{\boldsymbol{x}}\right)^2=\boldsymbol{a}_i'\boldsymbol{S}\boldsymbol{a}_i \xrightarrow{\text{approximate}} V\left(y_i\right)=\boldsymbol{a}_i'\boldsymbol{\Sigma}\boldsymbol{a}_i$$

then linear combination $\hat{y}_i=\boldsymbol{a}_i'\boldsymbol{x}$ is called the *i*-th Sample Principal Component, $i=1,2,\cdots,p$.

- Notes: Calculating sample principal component is to maximize sample variance not population variance.

§ 7.3.2 calculate principal component from $\boldsymbol{S}$

- Procedures:

step 1: Find eigenvalues of $\boldsymbol{S}$ $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p \geq 0$ and corresponding mutually orthogonal unit eigenvectors $\hat{\boldsymbol{t}}_1, \hat{\boldsymbol{t}}_2, \cdots, \hat{\boldsymbol{t}}_p$.

Step 2: Construct the i-th sample principal component $\hat{y}_i = \hat{\boldsymbol{t}}_i' \boldsymbol{x}$ which has the sample variance $\hat{\lambda}_i$, i=1,2,…,p, and

$$\mathrm{cov}(\hat{y}_i, \hat{y}_j) = 0 (i \neq j)$$

Notes: In geometry, orientations of p sample principal is the directions of $\hat{\boldsymbol{t}}_1, \hat{\boldsymbol{t}}_2, \cdots, \hat{\boldsymbol{t}}_p$. Projection points on $\hat{\boldsymbol{t}}_1$ of n samples are most dispersal. The dispersion degree of projection points of n samples on $\hat{\boldsymbol{t}}_2, \cdots, \hat{\boldsymbol{t}}_p$ is decreasing successively.

§ 7.3.2 calculate principal component from $\boldsymbol{S}$

Notice that $x_i = \hat{t}_{i1}\hat{y}_1 + \hat{t}_{i2}\hat{y}_2 + \ldots + \hat{t}_{ip}\hat{y}_p$ or $\mathbf{x} = \widehat{\boldsymbol{T}}\widehat{\boldsymbol{y}}$ with $\widehat{\boldsymbol{T}} = (\hat{t}_1, \hat{t}_2, \cdots, \hat{t}_p)$ and $\widehat{\boldsymbol{y}} = (\hat{y}_1, \hat{y}_2, \cdots, \hat{y}_p)'$

- Total sample variance:

$$\sum_{i=1}^{p} s_{ii} = \sum_{i=1}^{p} \hat{\lambda}_i \qquad (7.3.1)$$

  - $Tr(S) = Tr\left(\hat{T}'\hat{\Lambda}\hat{T}\right) = Tr\left(\hat{\Lambda}\hat{T}\hat{T}'\right) = Tr(\hat{\Lambda})$

- Sample correlation coefficients of x_i and $\hat{y}_k$:

$$Cov(x_i, \hat{y}_k) = \hat{\lambda}_k \hat{t}_{ik},\ r\left(x_i, \hat{y}_k\right) = \sqrt{\hat{\lambda}_k}\,\hat{t}_{ik} / \sqrt{s_{ii}},\quad i, k = 1, 2, \ldots, p \qquad (7.3.2)$$

where $\hat{\boldsymbol{t}}_k = \left(\hat{t}_{1k}, \hat{t}_{2k}, \cdots, \hat{t}_{pk}\right)'$, $k=1,2,\ldots,p.$

approximate

$$\rho(x_i, y_k) = \frac{\sqrt{\lambda_k}}{\sqrt{\sigma_{ii}}} t_{ik}, i, k = 12, \cdots, p \qquad (7.2.10)$$

Principal Component Score

- In real application, we usually minus $\boldsymbol{x}_j$ by $\overline{\boldsymbol{x}}$ to centralize samples. $\boldsymbol{S}$ will not change. The only change of the previous discussion is to rewrite *i*-th principal component as centralized form:

$$\hat{y}_i = \hat{\boldsymbol{t}}_i'\left(\boldsymbol{x} - \overline{\boldsymbol{x}}\right), \quad i = 1, 2, \cdots, p$$

- If observed vectors $\boldsymbol{x}$ is replaced by each observation value $\boldsymbol{x}_j$, then the *i*-th principal component

$$\hat{y}_{ji} = \hat{\boldsymbol{t}}_i'\left(\boldsymbol{x}_j - \overline{\boldsymbol{x}}\right), \quad i = 1, 2, \cdots, p, j = 1, 2, \ldots, n$$

is called as the *i*-th principal component score（主成分得分）of $\boldsymbol{x}_j$. Average principal component score of all observation value:

$$\overline{\hat{y}}_i = \frac{1}{n}\sum_{j=1}^{n}\hat{y}_{ji} = \frac{1}{n}\hat{\boldsymbol{t}}_i'\left(\sum_{j=1}^{n}\boldsymbol{x}_j - n\overline{\boldsymbol{x}}\right) = 0, \quad i = 1, 2, \cdots, p$$

§ 7.3.3 calculate principal component from $\hat{\boldsymbol{R}}$

- Suppose sample correlation matrix $\hat{\boldsymbol{R}}$'s p eigenvalues $\hat{\lambda}_1^* \geq \hat{\lambda}_2^* \geq \cdots \geq \hat{\lambda}_p^*$. $\hat{\boldsymbol{t}}_1^*, \hat{\boldsymbol{t}}_2^*, \cdots, \hat{\boldsymbol{t}}_p^*$ are corresponding orthogonal unit eigenvectors. Then i-th sample principal component:

$$\hat{y}_i^* = \hat{\boldsymbol{t}}_i^{*\prime} \boldsymbol{x}^*, \quad i = 1, 2, \cdots, p$$

where $\boldsymbol{x}^* = \hat{\boldsymbol{D}}^{-1}(\boldsymbol{x} - \bar{\boldsymbol{x}})$ is the standardized vector with

$\hat{\boldsymbol{D}} = \mathrm{diag}\left(\sqrt{s_{11}}, \sqrt{s_{22}}, \cdots, \sqrt{s_{pp}}\right)$ and $\hat{\boldsymbol{R}} = \hat{D}^{-1} S \hat{D}^{-1}$ $\xrightarrow{\text{approximate}}$ $\boldsymbol{R} = \mathrm{D}^{-1} \Sigma \mathrm{D}^{-1}$

Standardized Principal Component Score

- Let

$$\boldsymbol{x}_j^* = \hat{\boldsymbol{D}}^{-1}\left(\boldsymbol{x}_j - \bar{\boldsymbol{x}}\right)$$

It is the standardized data vector of $\boldsymbol{x}_j$. Replace $\boldsymbol{x}^*$ in the above formula by it. Then the i-th principal component score of observation value $\boldsymbol{x}_j$:

$$\hat{y}_{ji}^* = \{\hat{t}_i^*\}' x_j^*, i = 1,2,\cdots,p$$

Average principal component score of all the observation values:

$$\bar{\hat{y}}_i^* = \frac{1}{n}\sum_{j=1}^{n}\hat{y}_{ji}^* = \frac{1}{n}\hat{\boldsymbol{t}}_i^{*\prime}\sum_{j=1}^{n}\boldsymbol{x}_j^* = 0, \quad i = 1,2,\cdots,p$$

§ 7.3.4 application of principal components

- In principal component analysis, firstly we should make sure that the rate of accumulative contribution of the first several principal components reaches a high level. Secondly, the used principal components should have reasonable background and explanation.
- The explanation of principal components are not as clear and precise as that of original variables. This is the inevitable price should be paid in dimensional reduction. Therefore, number of principal components m should be obviously smaller than that of original variables p （unless p is small. Otherwise, the advantage of reducing dimension cannot be comparable to the advantage of clear explanation of original variables.

- If original variables have high correlation, then rate of accumulative contribution of a few principal components can attain at a high level. That is to say the rate of accumulative contribution is easy to satisfy.
- The difficulty of principal components is how to obtain a better explanation of principal components. If one of the selective principal components cannot be explained, the whole principal component analysis fails.
- Principal component analysis is a common way to reduce dimension. Simply speaking, in order to make full advantage of this method, it partially depends on reasonably selection and partially depends on "Luck".

Example 7.3.1 In the process of making clothing standard process, we measure 128 male adults with the following 6 indexes: height(x_1), sitting height(x_2), chest measurement (x_3), arm length(x_4), rib circumstance(x_5) and waistline(x_6). Sample correlation matrix is listed in Table 7.3.1. Table 7.3.1 Sample correlation matrix

	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1.000					
x_2	0.79	1.000				
x_3	0.36	0.31	1.000			
x_4	0.76	0.55	0.35	1.000		
x_5	0.25	0.17	0.64	0.16	1.000	
x_6	0.51	0.35	0.58	0.38	0.63	1.000

- First three eigenvalues, eigenvectors and rates of contribution of correlation matrix $\hat{\boldsymbol{R}}$ are listed in Table 7.3.2.

Table 7.3.2 eigenvalues, eigenvectors and rates of contribution of $\hat{\boldsymbol{R}}$

Eigenvectors	$\hat{\boldsymbol{t}}_1$	$\hat{\boldsymbol{t}}_2$	$\hat{\boldsymbol{t}}_3$
x_1^*(身高)	0.469	−0.365	0.092
x_2^*(坐高)	0.404	−0.397	0.613
x_3^* (胸围)	0.394	0.397	−0.279
x_4^* (上臂长)	0.408	−0.365	−0.705
x_5^*(肋围)	0.337	0.569	0.164
x_6^* (腰围)	0.427	0.308	0.119
Eigenvalue	3.287	1.406	0.459
Rate of contribution	0.548	0.234	0.077
Rate of accumulative contribution	0.548	0.782	0.859

- First three principal components:

$$\hat{y}_1 = 0.469x_1^* + 0.404x_2^* + 0.394x_3^* + 0.408x_4^* + 0.337x_5^* + 0.427x_6^*$$

$$\hat{y}_2 = -0.365x_1^* - 0.397x_2^* + 0.397x_3^* - 0.365x_4^* + 0.569x_5^* + 0.308x_6^*$$

$$\hat{y}_3 = 0.092x_1^* + 0.613x_2^* - 0.279x_3^* - 0.705x_4^* + 0.164x_5^* + 0.119x_6^*$$

- From Table 7.3.2, rate of accumulative contribution of first two principal components has reached to 78.2%. That of three is 85.9 %. Therefore, the first two or three principal components have already represented original variables.
- First principal component $\hat{y}_1$ has approximately equal positive loading to each (standardized) original variable. So, it is called size component(身材大小成分).

- Second principal component $\hat{y}_2$ has medium negative loading on x_1^*, x_2^*, x_4^* and medium positive loading on x_3^*, x_5^*, x_6^*. It is called shape component （or fat and thin component） .
- Third principal component $\hat{y}_3$ has big positive loading on x_2^* and big negative loading on x_4^* . It has smaller loadings on other variables. It is called brachium component（臂长成分） .
- Since the rate of contribution of third principal component is not high(7.65%) and has not important real meaning. Therefore, we consider to take the first two principal components
- Since $\hat{\lambda}_6$ is very small, there exists collinearity relationship :

$$\hat{\lambda}_6 = 0.126, \hat{\boldsymbol{t}}_6 = (-0.786, 0.433, -0.125, 0.371, 0.034, 0.179)'$$

$$-0.786x_1^* + 0.443x_2^* - 0.125x_3^* + 0.371x_4^* + 0.034x_5^* + 0.179x_6^* \approx 0$$

$$\hat{y}_6 = t_6' x^* \quad \mathrm{var}(\hat{y}_6) = 0.126 \approx 0$$

- Example 7.3.2 In exercise 6.5, 8 items of man track sport （男子径赛运动） record：

x_1： 100m （s） x_5： 1500m （s）

x_2： 200m （s） x_6： 5000m （s）

x_3： 400m （s） x_7： 10000m （s）

x_4： 800m （s） x_8： marathon （m）

Table 7.3.3 Sample Correlation Matrix

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
x_1	1.000							
x_2	0.923	1.000						
x_3	0.841	0.851	1.000					
x_4	0.756	0.807	0.870	1.000				
x_5	0.700	0.775	0.835	0.918	1.000			
x_6	0.619	0.695	0.779	0.864	0.928	1.000		
x_7	0.633	0.697	0.787	0.869	0.935	0.975	1.000	
x_8	0.520	0.596	0.705	0.806	0.866	0.932	0.943	1.000

Table 7.3.4 First three eigenvalues, eigenvectors and rates of contribution of $\hat{R}$

Eigenvectors	$\hat{t}_1$	$\hat{t}_2$	$\hat{t}_3$
x_1^*: 100m	0.318	0.567	0.332
x_2^*: 200m	0.337	0.462	0.361
x_3^*: 400m	0.356	0.248	−0.560
x_4^*: 800m	0.369	0.012	−0.532
x_5^*: 1500m	0.373	−0.140	−0.153
x_6^*: 5000m	0.364	−0.312	0.190
x_7^*: 10000m	0.367	−0.307	0.182
x_8^*: marathon	0.342	−0.439	0.263
Eigenvalues	6.622	0.878	0.159
Contribution rate	0.828	0.110	0.020
Accumulative contribution rate	0.828	0.937	0.957

Example 7.3.2

The following are eight men's track and field event records:

x_1: 100 meters (seconds), x_2: 200 meters (seconds), x_3: 400 meters (seconds),

x_4: 800 meters (seconds), x_5 : 1500 meters (minutes), x_6 : 5000 meters (minutes),

x_7 : 10000 meters (minutes), x_8 : Marathon (minutes).

The sample correlation matrix of $x_1, x_2, \cdots, x_8$ is listed in Table 7.3.3. From this table, an interesting phenomenon can be observed in the correlation coefficients of the eight variables: the correlation coefficient of x_1 with $x_2, x_3, \cdots, x_8$ decreases in sequence, similarly, the correlation coefficient of x_2 with $x_3, x_4, \cdots, x_8$ also decreases in sequence, and so on. The results of the calculation of the eigenvalues, eigenvectors, and contribution rates of the sample correlation matrix $\widehat{R}$ are listed in Table 7.3.4.

Example 7.3.2

Table 7.3.4. The first three eigenvalues, eigenvectors, and their contribution rates of $\hat{R}$.

特征向量	$\hat{t}_1$	$\hat{t}_2$	$\hat{t}_3$
x_1^* :100 米	0.318	0.567	0.332
x_2^* :200 米	0.337	0.462	0.361
x_3^* :400 米	0.356	0.248	−0.560
x_4^* :800 米	0.369	0.012	−0.532
x_5^* :1 500 米	0.373	−0.140	−0.153
x_6^* :5 000 米	0.364	−0.312	0.190
x_7^* :10 000 米	0.367	−0.307	0.182
x_8^* :马拉松	0.342	−0.439	0.263
特征值	6.622	0.878	0.159
贡献率	0.828	0.110	0.020
累计贡献率	0.828	0.937	0.957

- The cumulative contribution rate of the first two principal components has reached 93.7%, nearly explaining the entire total variance.
- $\hat{y}_1$: strong and weak elements in track events（田径赛项目）.
- $\hat{y}_2$: reflects the contrast between speed and endurance.

- Example 6.3.3. Table 6.3.10 presents data on eight key variables related to the average annual per capita consumption expenditure of urban households （城镇居民家庭平均每人全年消费性支出） across 31 provinces, municipalities, and autonomous regions in China for the year 1999. These variables are respectively:

x_1: food x_5: transportation and communication

x_2: clothing x_6: entertainment, education and culture

x_3: family equipment and service x_7: settlement

x_4: health care x_8: sundry commodity and service

- Separately use shortest distance method, centroid method and Ward method to do cluster analysis of data from every area. Before the cluster analysis, standardize all of the variables.

Table 6.3.14 Consumer Spending Data Unit: yuan

Area	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
北京	2959.19	730.79	749.41	513.34	467.87	1141.82	478.42	457.64
天津	2459.77	495.47	697.33	302.87	284.19	735.97	570.84	305.08
河北	1495.63	515.9	362.37	285.32	272.95	540.58	364.91	188.63
山西	1406.33	477.77	290.15	208.57	201.5	414.72	281.84	212.1
内蒙古	1303.97	524.29	254.83	192.17	249.81	463.09	287.87	192.96
辽宁	1730.84	553.9	246.91	279.81	239.18	445.2	330.24	163.86
吉林	1561.86	492.42	200.49	218.36	220.69	459.62	360.48	147.76
黑龙江	1410.11	510.71	211.88	277.11	224.65	376.82	317.61	152.85
上海	3712.31	550.74	893.37	346.93	527	1034.98	720.33	462.03
江苏	2207.58	449.37	572.4	211.92	302.09	585.23	429.77	252.54
浙江	2629.16	557.32	689.73	435.69	514.66	795.87	575.76	323.36
安徽	1844.78	430.29	271.28	126.33	250.56	513.18	314	151.39
福建	2709.46	428.11	334.12	160.77	405.14	461.67	535.13	232.29
江西	1563.78	303.65	233.81	107.9	209.7	393.99	509.39	160.12
山东	1675.75	613.32	550.71	219.79	272.59	599.43	371.62	211.84

河南	1427.65	431.79	288.55	208.14	217	337.76	421.31	165.32
湖北	1783.43	511.88	282.84	201.01	237.6	617.74	523.52	182.52
湖南	1942.23	512.27	401.39	206.06	321.29	697.22	492.6	226.45
广东	3055.17	353.23	564.56	356.27	811.88	873.06	1082.82	420.81
广西	2033.87	300.82	338.65	157.78	329.06	621.74	587.02	218.27
海南	2057.86	186.44	202.72	171.79	329.65	477.17	312.93	279.19
重庆	2303.29	589.99	516.21	236.55	403.92	730.05	438.41	225.8
四川	1974.28	507.76	344.79	203.21	240.24	575.1	430.36	223.46
贵州	1673.82	437.75	461.61	153.32	254.66	445.59	346.11	191.48
云南	2194.25	537.01	369.07	249.54	290.84	561.91	407.7	330.95
西藏	2646.61	839.7	204.44	209.11	379.3	371.04	269.59	389.33
陕西	1472.95	390.89	447.95	259.51	230.61	490.9	469.1	191.34
甘肃	1525.57	472.98	328.9	219.86	206.65	449.69	249.66	228.19
青海	1654.69	437.77	258.78	303	244.93	479.53	288.56	236.51
宁夏	1375.46	480.89	273.84	317.32	251.08	424.75	228.73	195.93
新疆	1608.82	536.05	432.46	235.82	250.28	541.3	344.85	214.4

- Example 7.3.3 Do principal component analysis based on correlation matrix $\hat{\boldsymbol{R}}$ to example 6.3.3. After calculation, sample correlation matrix $\hat{\boldsymbol{R}}$ of $x_1, x_2, \cdots, x_8$ are listed in Table 7.3.5. First three eigenvalues, eigenvectors and rates of contribution of $\hat{\boldsymbol{R}}$ are listed in Table 7.3.6.

Table 7.3.5 Sample Correlation Matrix

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
x_1	1.000							
x_2	0.247	1.000						
x_3	0.698	0.258	1.000					
x_4	0.468	0.423	0.621	1.000				
x_5	0.828	0.086	0.585	0.531	1.000			
x_6	0.769	0.255	0.856	0.684	0.708	1.000		
x_7	0.670	−0.201	0.569	0.314	0.800	0.647	1.000	
x_8	0.877	0.349	0.667	0.628	0.776	0.745	0.525	1.000

Table7.3.6 First three eigenvalues, eigenvectors and rates of contribution of $\hat{R}$

Eigenvectors	$\hat{t}_1$	$\hat{t}_2$	$\hat{t}_3$
x_1^*: food	0.401	−0.077	0.415
x_2^*: clothing	0.132	0.749	0.332
x_3^*: family equipment and service	0.375	0.065	−0.442
x_4^*: health care	0.320	0.345	−0.478
x_5^*: transportation and communication	0.388	−0.232	0.279
x_6^*: entertainment and education service	0.406	0.027	−0.310
x_7^*: settlement	0.326	−0.496	−0.034
x_8^*: sundry commodity and service	0.396	0.096	0.345
eigenvalue	5.098	1.352	0.574
Contribution rate	0.637	0.169	0.072
Accumulative contribution rate	0.637	0.806	0.878

第一个主成分解释：综合性消费支出成分

第二个主成分解释：消费倾向成分

- First principle component: comprehensive consumption expenditure(综合消费性支出)
- Second principle component: Propensity （倾向性）consumption expenditure

Table 7.3.7 Ranked by first principal component for 31 regions

Region	$\hat{y}_1$	$\hat{y}_2$	Region	$\hat{y}_1$	$\hat{y}_2$
江西	−2.234	−1.867	新疆	−0.697	0.647
河南	−1.947	−0.388	四川	−0.533	0.041
黑龙江	−1.927	0.636	广西	−0.251	−2.058
吉林	−1.859	0.151	山东	−0.147	0.983
山西	−1.848	0.404	福建	0.201	−1.337
内蒙古	−1.826	0.509	湖南	0.219	−0.203
安徽	−1.796	−0.519	江苏	0.407	−0.311
甘肃	−1.549	0.526	云南	0.435	0.479
宁夏	−1.501	0.906	西藏	0.437	2.365
辽宁	−1.313	0.844	重庆	1.115	0.409
贵州	−1.298	−0.341	天津	2.006	0.044
海南	−1.157	−1.913	浙江	3.583	0.531
青海	−1.045	0.426	北京	5.426	2.466
陕西	−0.859	−0.501	广东	5.583	−3.072
河北	−0.769	0.580	上海	5.866	−0.195
湖北	−0.717	−0.247			

第一个主成分解释：综合性消费支出成分

Table 7.3.8 Ranked by second principal component for 31 regions

Region	$\hat{y}_1$	$\hat{y}_2$	Region	$\hat{y}_1$	$\hat{y}_2$
广东	5.583	−3.072	山西	−1.848	0.404
广西	−0.251	−2.058	重庆	1.115	0.409
海南	−1.157	−1.913	青海	−1.045	0.426
江西	−2.234	−1.867	云南	0.435	0.479
福建	0.201	−1.337	内蒙古	−1.826	0.509
安徽	−1.796	−0.519	甘肃	−1.549	0.526
陕西	−0.859	−0.501	浙江	3.583	0.531
河南	−1.947	−0.388	河北	−0.769	0.580
贵州	−1.298	−0.341	黑龙江	−1.927	0.636
江苏	0.407	−0.311	新疆	−0.697	0.647
湖北	−0.717	−0.247	辽宁	−1.313	0.844
湖南	0.219	−0.203	宁夏	−1.501	0.906
上海	5.866	−0.195	山东	−0.147	0.983
四川	−0.533	0.041	西藏	0.437	2.365
天津	2.006	0.044	北京	5.426	2.466
吉林	−1.859	0.151			

第二个主成分解释：消费倾向成分

南低北高

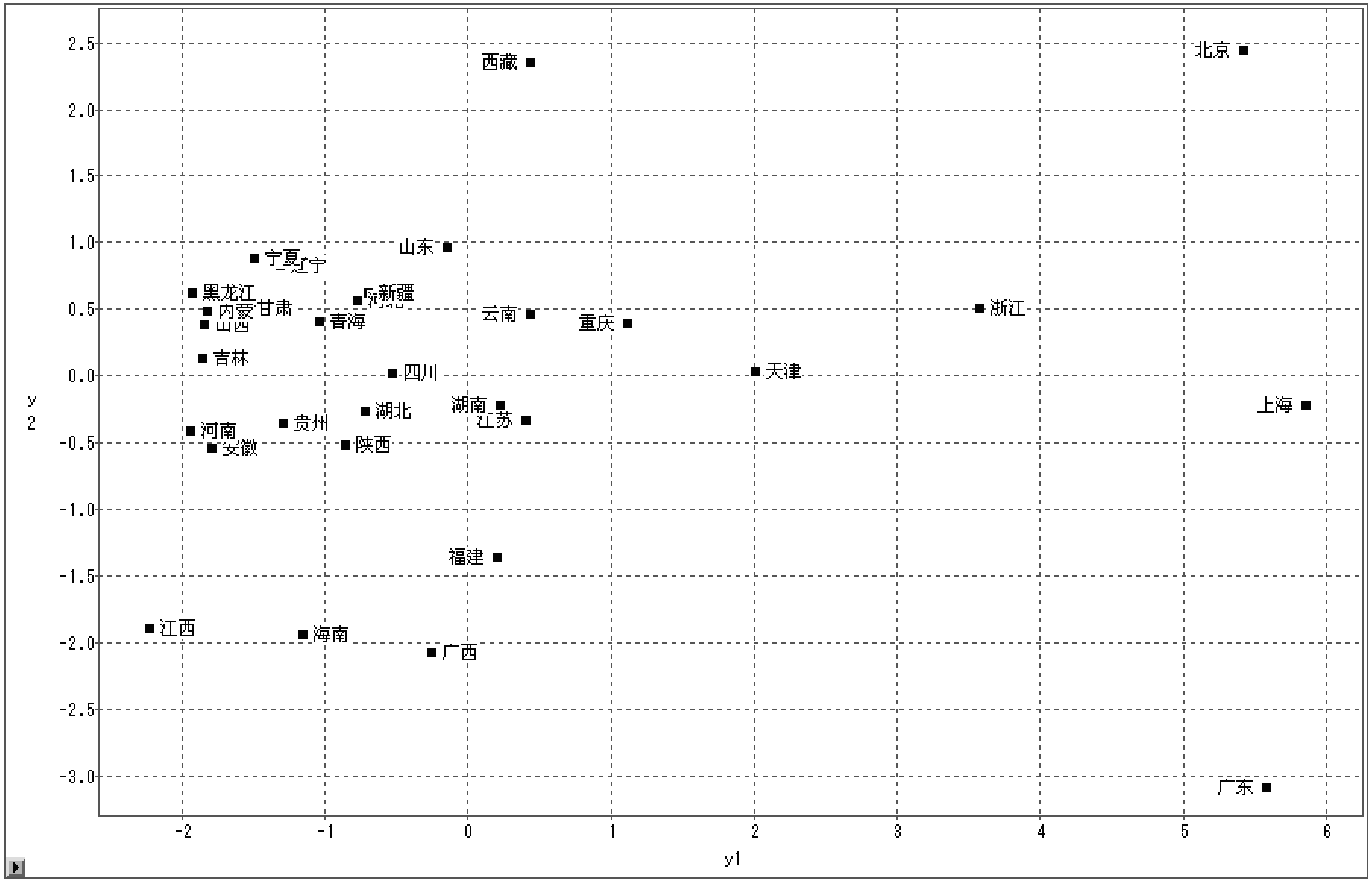
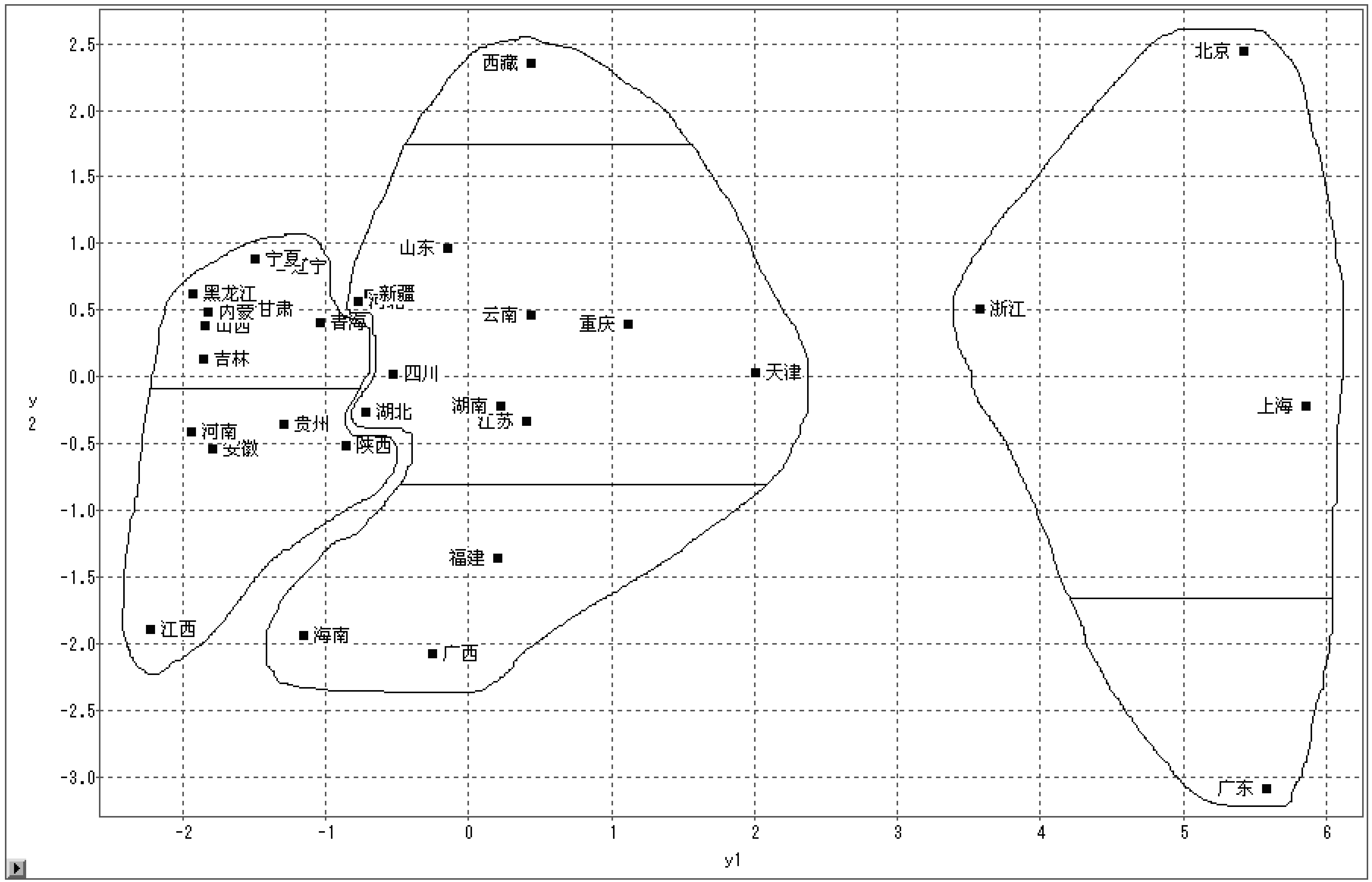


Figure 7.3.3 The scatter diagram of two principal components

Some remarks:

- i) Shanghai, Guangdong and Beijing are on the far right, and the comprehensive consumption expenditure of urban residents is the highest;
- ii) Beijing and Tibet are at the top of the scatter chart, indicating that the consumption expenditure affected by the regional climate accounts for the highest proportion in these regions;
- iii) Guangdong is at the bottom of the scatter chart, indicating that the proportion of consumption expenditure affected by regional climate is the lowest.

Whether the clustering using principal components is consistent with the systematic clustering directly starting from the original variables. In order to obtain perceptual knowledge, let's see how the clustering result of the sum of squares of deviation method in example 6.3.3 (see figure 6.3.14) appears in figure 7.3.3



使用离差平方和方法的聚类结果（见图6.3.314）在图7.3.3上的显示

Assignment： see the assignment files